Variance and Covariance

math-foundations / 26-variance-covariance

1 Setup

Throughout this lesson, fix a probability space $(\Omega, \mathcal{F}, P)$ and let $L^2(P) = \{X : \Omega \to \mathbb{R} \mid \mathbb{E}[X^2] < \infty\}$ denote the Hilbert space of square-integrable random variables. All random variables below are assumed to lie in $L^2(P)$, which guarantees the existence of the moments $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and (by Cauchy–Schwarz) $\mathbb{E}[XY]$ for $X, Y \in L^2(P)$.

2 Definitions

Definition 1 (Variance)**.** The *variance* of $X \in L^2(P)$ is

$$\mathrm{Var}(X) \;\coloneqq\; \mathbb{E}\big[(X - \mathbb{E}[X])^2\big]\,.$$

By expanding the square and applying linearity of expectation,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The *standard deviation* is $\sigma_X = \sqrt{\mathrm{Var}(X)}$.

Definition 2 (Covariance)**.** The *covariance* of $X, Y \in L^2(P)$ is

$$\mathrm{Cov}(X, Y) \;\coloneqq\; \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \;=\; \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y].$$

Note $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.

Definition 3 (Correlation)**.** If $\sigma_X, \sigma_Y > 0$, the *(Pearson) correlation coefficient* of X and Y is

$$\rho_{X,Y} \;\coloneqq\; \frac{\mathrm{Cov}(X, Y)}{\sigma_X\,\sigma_Y}.$$

By Cauchy–Schwarz applied to the centered variables, $\rho_{X,Y} \in [-1, 1]$.

3 Elementary properties

Lemma 1 (Bilinearity and symmetry of covariance)**.** *For $a, b, c \in \mathbb{R}$ and $X, Y, Z \in L^2(P)$:*

1. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$.
2. $\mathrm{Cov}(aX + b, Y) = a\ \mathrm{Cov}(X, Y)$.
3. $\mathrm{Cov}(X + Z, Y) = \mathrm{Cov}(X, Y) + \mathrm{Cov}(Z, Y)$.

Corollary 2 (Affine scaling of variance)**.** *For $a, b \in \mathbb{R}$, $\mathrm{Var}(aX + b) = a^2\ \mathrm{Var}(X)$.*

4 Variance of a sum (main theorem)

Theorem 3 (Variance of a sum)**.** *For $X, Y \in L^2(P)$,*

$$\operatorname{Var}(X+Y) \;=\; \operatorname{Var}(X) \;+\; \operatorname{Var}(Y) \;+\; 2\operatorname{Cov}(X,Y).$$

Proof. Let $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. By linearity of expectation, $\mathbb{E}[X+Y] = \mu_X + \mu_Y$. Applying the definition of variance to $X+Y$,

$$\begin{aligned}
\operatorname{Var}(X+Y) &= \mathbb{E}\Big[\,\big((X+Y) - \mathbb{E}[X+Y]\big)^2\,\Big] \\
&= \mathbb{E}\Big[\,\big((X-\mu_X) + (Y-\mu_Y)\big)^2\,\Big] \\
&= \mathbb{E}\big[\,(X-\mu_X)^2 + 2(X-\mu_X)(Y-\mu_Y) + (Y-\mu_Y)^2\,\big].
\end{aligned}$$

By linearity of expectation,

$$\begin{aligned}
\operatorname{Var}(X+Y) &= \mathbb{E}\big[(X-\mu_X)^2\big] + 2\,\mathbb{E}[(X-\mu_X)(Y-\mu_Y)] + \mathbb{E}\big[(Y-\mu_Y)^2\big] \\
&= \operatorname{Var}(X) + 2\operatorname{Cov}(X,Y) + \operatorname{Var}(Y),
\end{aligned}$$

where the last equality uses the definitions of variance and covariance. □

Corollary 4 (n-term version)**.** *For $X_1, \ldots, X_n \in L^2(P)$,*

$$\operatorname{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \operatorname{Var}(X_i) + 2\sum_{1\le i<j\le n} \operatorname{Cov}(X_i, X_j).$$

In particular, if the X_i are pairwise uncorrelated, then $\operatorname{Var}(\sum X_i) = \sum \operatorname{Var}(X_i)$.

Remark. Independence of X and Y implies $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, hence $\operatorname{Cov}(X,Y) = 0$. The converse is false: take $X \sim \operatorname{Uniform}(\{-1,0,1\})$ and $Y = X^2$; then $\operatorname{Cov}(X,Y) = 0$ but Y is a deterministic function of X.

5 Worked example

Bernoulli. Let $X \sim \operatorname{Bernoulli}(p)$. Then $\mathbb{E}[X] = p$, $\mathbb{E}[X^2] = p$, so

$$\operatorname{Var}(X) = p - p^2 = p(1-p).$$

The variance is maximized at $p = 1/2$.